

DS Unit 1 Previous Year Questions

Data Structures (DS) · Unit 1 · Arrays, Linked Lists & Complexity · 2025

Questions collected from previous year university examinations (2021–2025).

Part A — Short answer (2 marks)

1. Discuss the role of Asymptotic Notation in Arrays, Linked Lists & Complexity. [2025]
2. Compare Arrays with Polynomial Representation. When would you prefer one over the other? [2022]
3. Discuss the role of Singly Linked Lists in Arrays, Linked Lists & Complexity. [2024]
4. Illustrate Doubly and Circular Linked Lists with a worked example. [2024]

Part B — Long answer (10 marks)

1. Define Sparse Matrices. Derive or justify the key properties involved. [2023]
2. Illustrate Polynomial Representation with a worked example. [2024]
3. Define Asymptotic Notation. Derive or justify the key properties involved. [2022]
4. Compare Arrays with Singly Linked Lists. When would you prefer one over the other? [2023]
5. Discuss the role of Singly Linked Lists in Arrays, Linked Lists & Complexity. [2023]
6. Define Doubly and Circular Linked Lists. Derive or justify the key properties involved. [2021]
7. Illustrate Sparse Matrices with a worked example. [2025]
8. Define Polynomial Representation. Derive or justify the key properties involved. [2022]
9. Define Asymptotic Notation. Derive or justify the key properties involved. [2023]
10. Illustrate Arrays with a worked example. [2022]